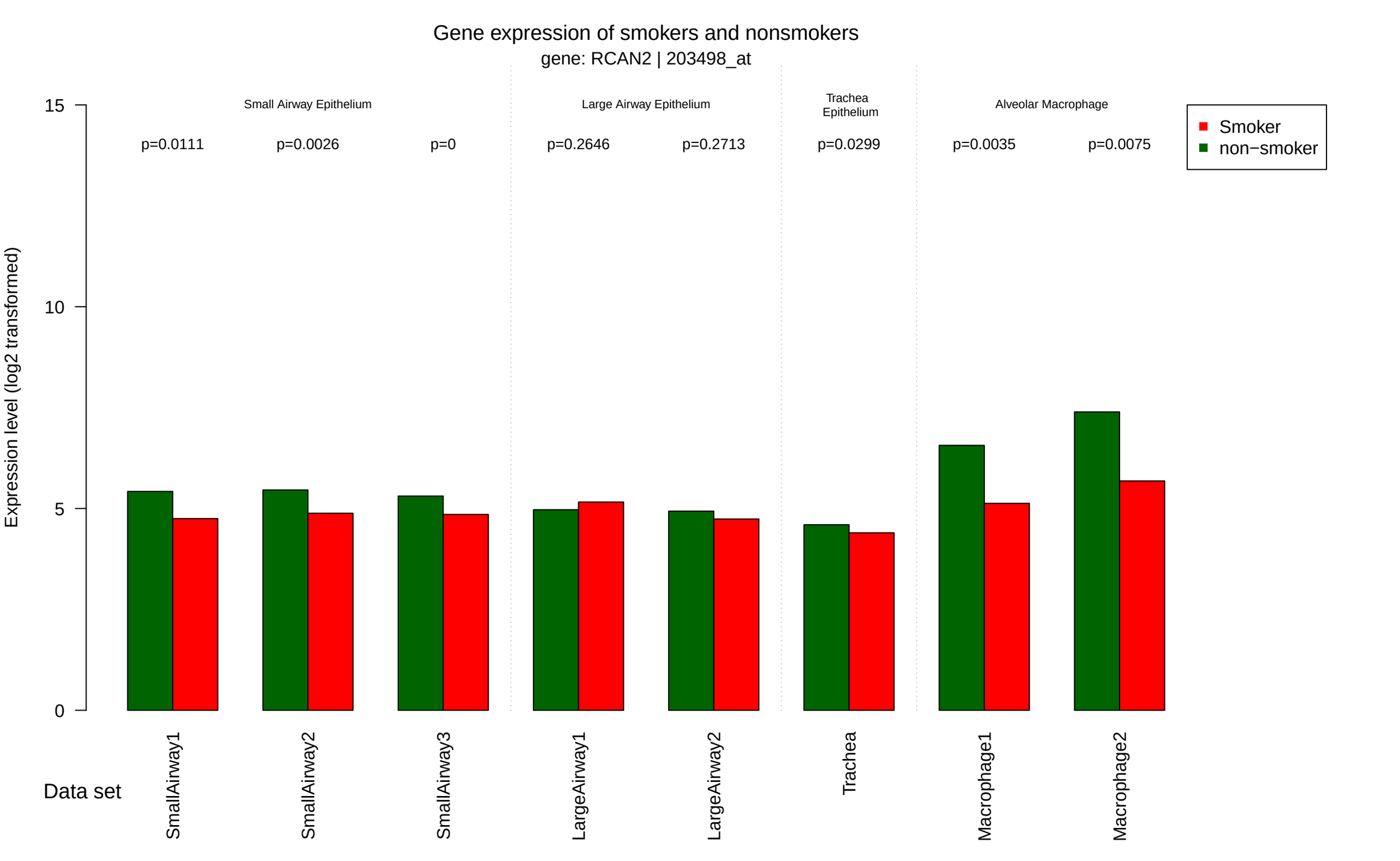
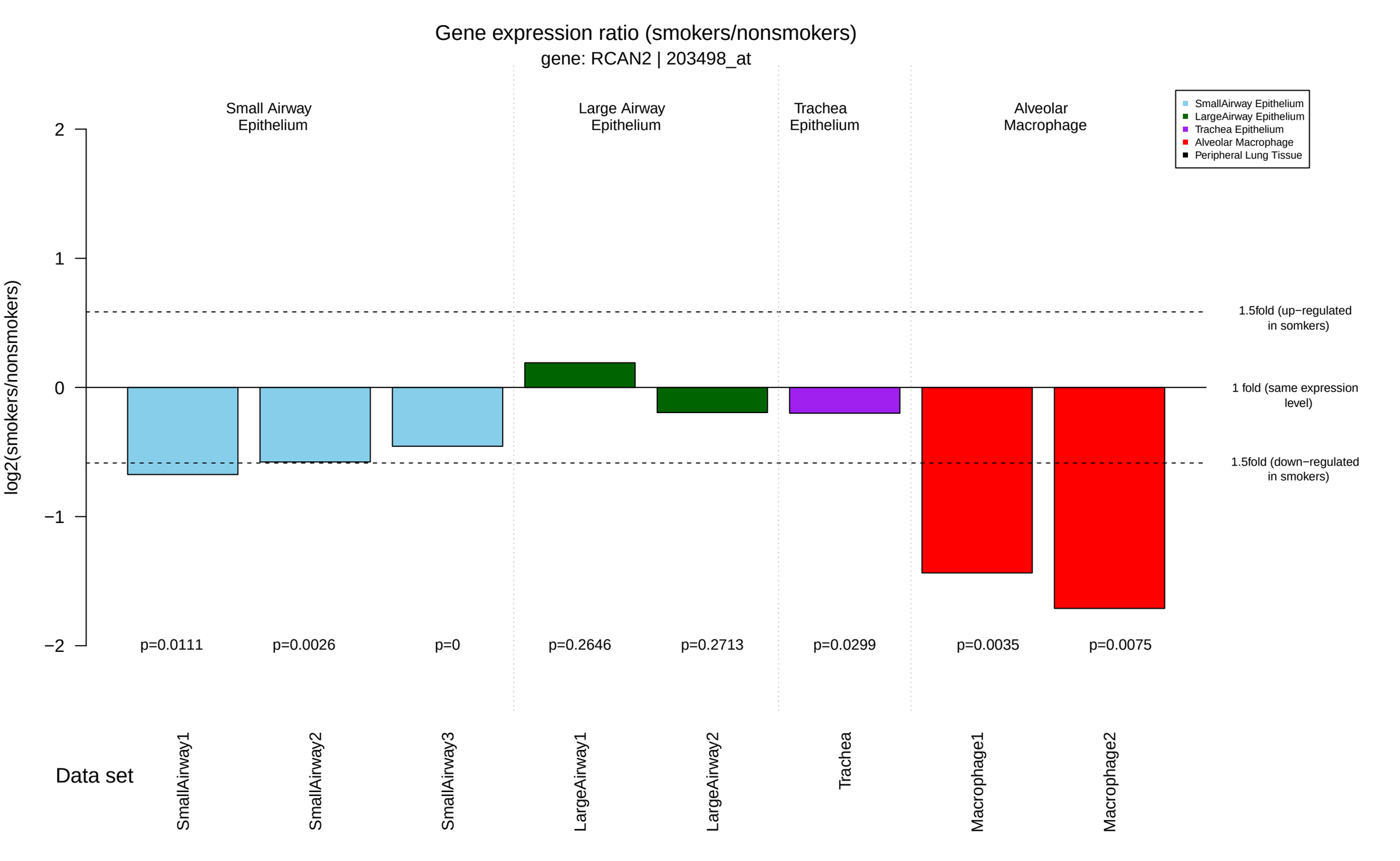
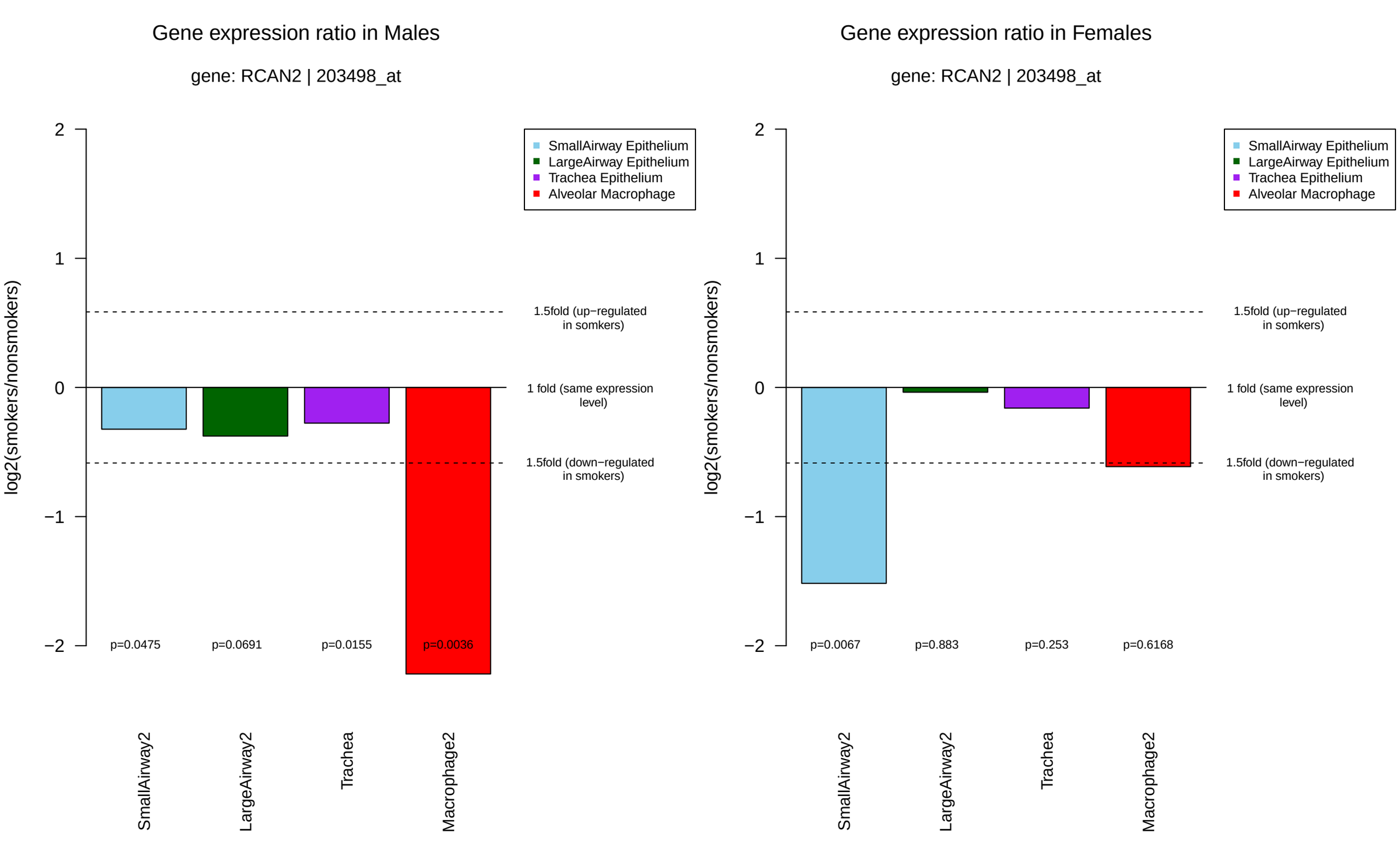
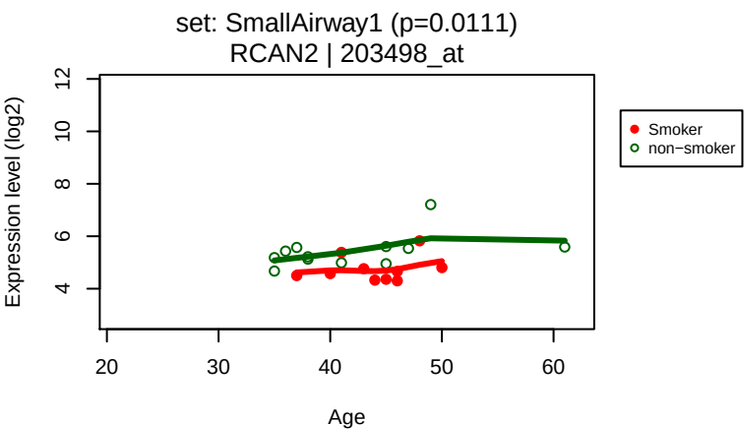
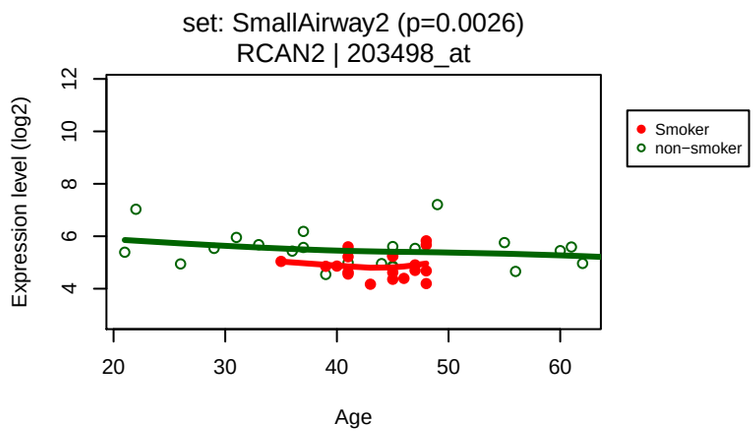
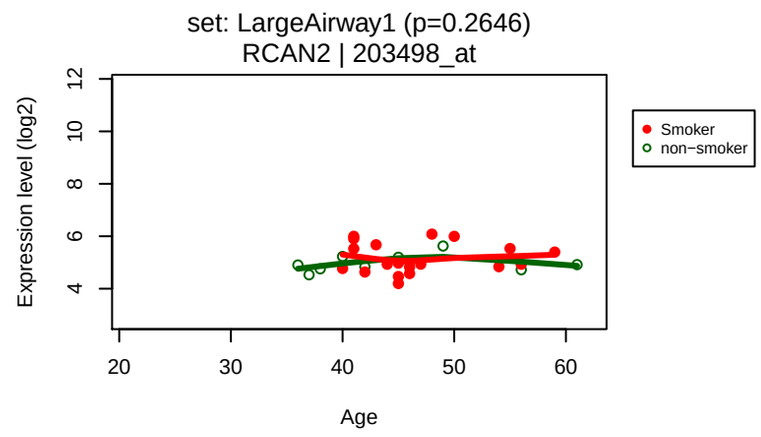

Gene expression of smokers and nonsmokers
gene: RCAN2 | 203498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0111
p=0.0026
p=0
p=0.2646
p=0.2713
p=0.0299
p=0.0035
p=0.0075
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

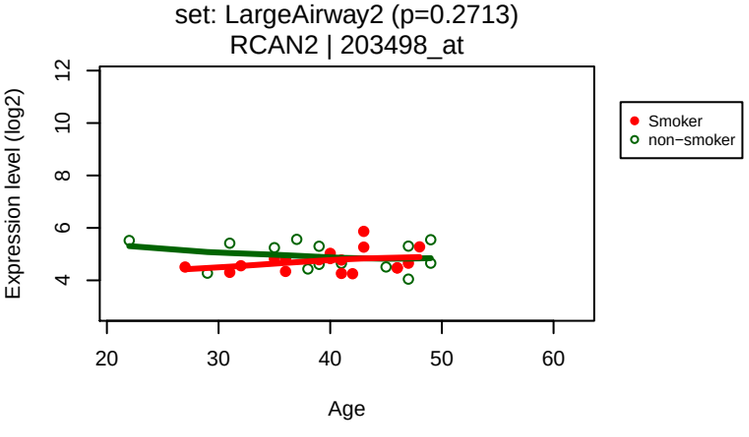

Gene expression ratio (smokers/nonsmokers)
gene: RCAN2 | 203498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0111
p=0.0026
p=0
p=0.2646
p=0.2713
p=0.0299
p=0.0035
p=0.0075
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

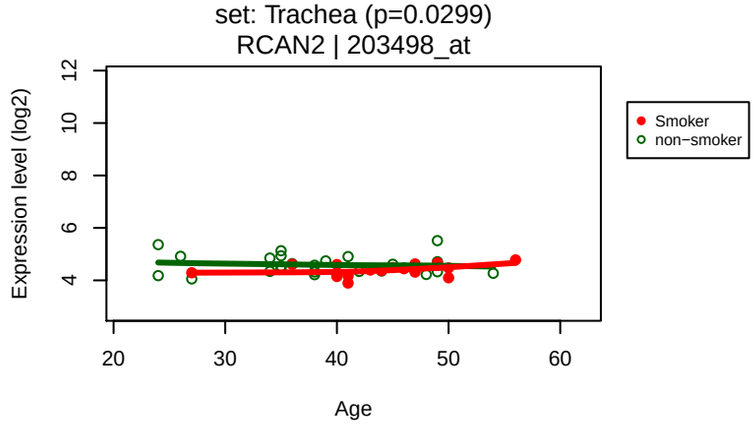

Gene expression ratio in Males
gene: RCAN2 | 203498_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0475
p=0.0691
p=0.0155
p=0.0036
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCAN2 | 203498_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0067
p=0.883
p=0.253
p=0.6168
SmallAirway2
LargeAirway2
Trachea
Macrophage2

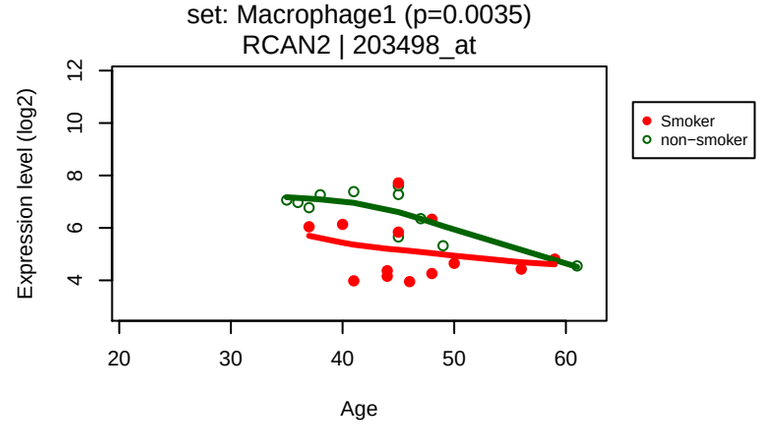

set: SmallAirway1 (p=0.0111)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker

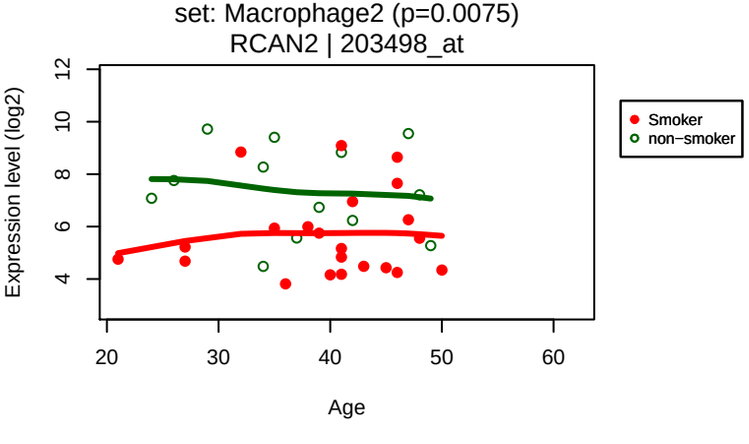

set: SmallAirway2 (p=0.0026)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker

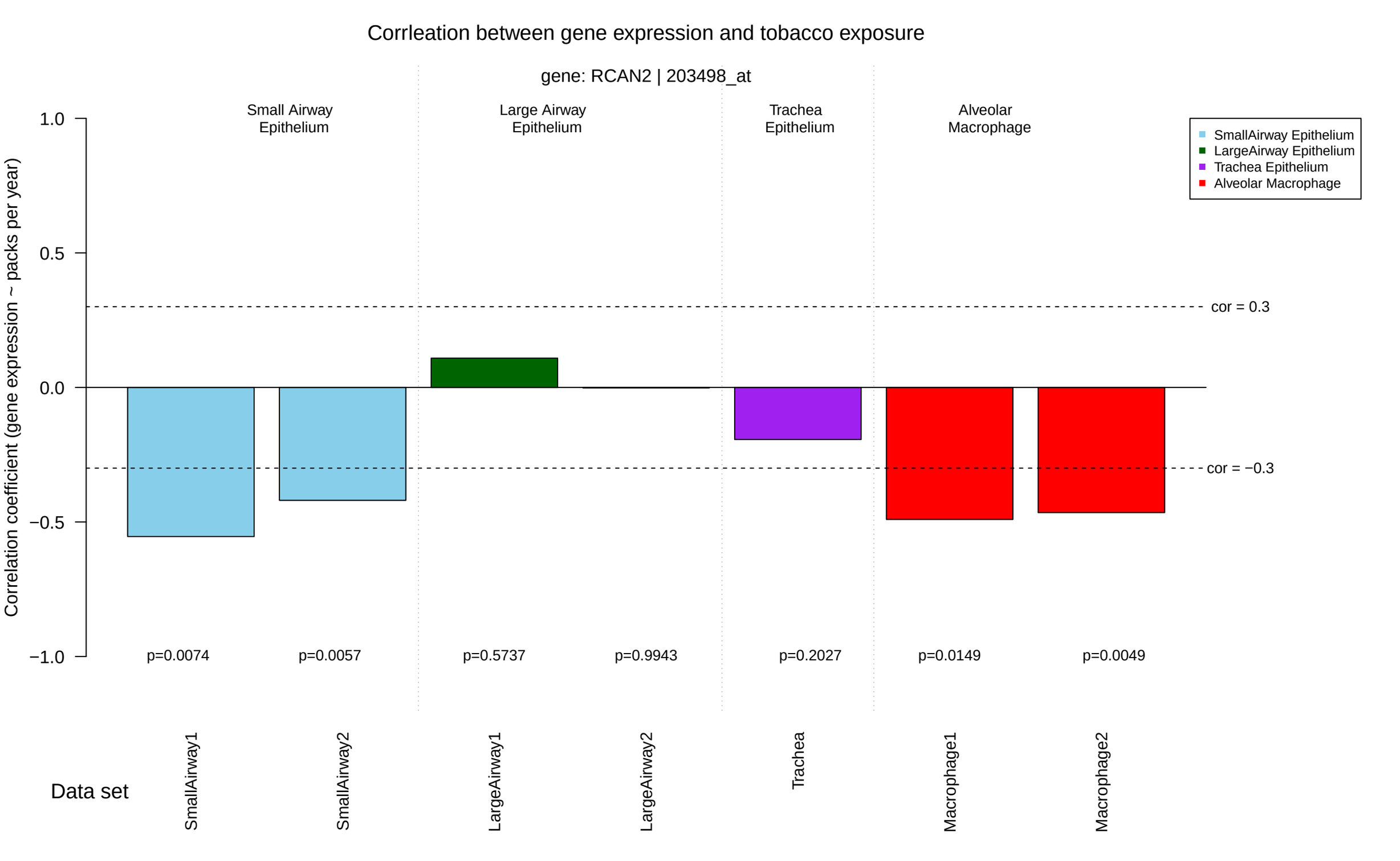

set: LargeAirway1 (p=0.2646)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2713)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0299)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0035)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0075)
RCAN2 | 203498_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCAN2 | 203498_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0074
p=0.0057
p=0.5737
p=0.9943
p=0.2027
p=0.0149
p=0.0049
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2